

Memo 02: Reconstructed Delaware River Test Scenarios

Codex review for FVCOM-MP test-case recovery

April 24, 2026

1 Purpose

This memo reconstructs the intent of the three Delaware River test cases stored in `FVCOM_MP_test_run`: cases **a**, **b**, and **c**. The goal is to recover what is shared across the setups, what actually differs among them, and what those differences imply for later mass-conservation testing of the flocculation-enabled plastic model.

2 Recovered Test Bundle

2.1 Shared Domain and Forcing Files

The common input bundle in `FVCOM_MP_test_run/INPUT` defines a single Delaware River Estuary configuration with:

- 68416 mesh nodes and 30 sigma layers from `waterPACT_DRE_its.nc`;
- 100 open-boundary nodes in `waterPACT_obc_2019.nc` and `waterPACT_tsobc_2019.nc`;
- 9 river entries in `waterPACT_riv_floc_MP.nml`: 6 Delaware River points (`DR_1`–`DR_6`) and 3 Schuylkill River points (`SR_1`–`SR_3`);
- hourly wind and pressure forcing for calendar year 2019;
- a river forcing file that spans 2019/01/01 00:00:01 to 2020/01/01 00:00:01.

The model runs themselves are set from modified Julian day 58485.0 to 58848.0, i.e. 2019-01-02 00:00:00 through 2019-12-31 00:00:00. Therefore the shared forcing files provide roughly a one-day pad before the model start.

2.2 River Source Characteristics

The river forcing file `waterPACT_riv_floc_MP.nc` contains:

- `river_flux` in $\text{m}^3 \text{s}^{-1}$;
- `coarse_sand_1` in g L^{-1} ;
- `mp1` in kg m^{-3} .

Two points are especially clear:

1. `mp1` is constant in time and identical for all 9 rivers:

$$\text{mp1} = 1.754906975 \times 10^{-6} \text{ kg m}^{-3}.$$

2. `coarse_sand_1` varies in time, with domain-wide minimum and maximum values

$$4.516 \times 10^{-4} \text{ g L}^{-1} \leq \text{coarse_sand_1} \leq 1.6097 \times 10^{-2} \text{ g L}^{-1}.$$

Because the plastic concentration is constant, the imposed plastic source differs among rivers only through the discharge `river_flux`. Using the hourly records in `Times`, the implied annual river-source totals are approximately:

- Delaware group: $1.379 \times 10^{10} \text{ m}^3$ water, $2.420 \times 10^4 \text{ kg}$ plastic, $6.578 \times 10^7 \text{ kg}$ sediment;
- Schuylkill group: $3.915 \times 10^9 \text{ m}^3$ water, $6.870 \times 10^3 \text{ kg}$ plastic, $1.351 \times 10^7 \text{ kg}$ sediment.

3 Case-by-Case Reconstruction

Case	Intended role recovered from files	Plastic floc switch	Sediment setting	Other notable differences
a	Baseline plastic transport case without active plastic-sediment floc interaction.	<code>PLAST_FLOC = F</code> in <code>generic_plastic_and_sediment</code>	<code>SEDIMENT_MODEL = F</code> . The <code>sediment</code> file name is still present in the namelist but should be inactive.	Uses <code>STARTUP_TS_TYPE = 'observed'</code> from <code>waterPACT_DRE_its.nc</code> . NetCDF and restart output begin on MJD 58490.0, i.e. 5 days after the model start.
b	Main floc-enabled Delaware test with active suspended sediment and river plastic input.	<code>PLAST_FLOC = T</code> in <code>generic_plastic_and_sediment</code>	<code>SEDIMENT_MODEL = T</code> with <code>generic_sediment</code> as the only sediment class is <code>coarse_sand_1</code> with <code>SED_SD50 = 0.1 mm</code> .	Uses <code>STARTUP_TS_TYPE = 'constant'</code> with constant $T = 20^\circ\text{C}$ and $S = 30 \text{ PSU}$. NetCDF and restart output begin at the model start.

Case	Intended role recovered from files	Plastic floc switch	Sediment setting	Other notable differences
c	Sensitivity test relative to case b, most likely aimed at changing sediment character while keeping the plastic settings fixed.	PLAST_FLOC = T. <code>generic_plastic_c.inp</code> is byte-for-byte identical to <code>generic_plastic_b.inp</code>	SEDIMENT_MODEL = <code>inf</code> with <code>generic_sediment_c.inp</code> . The only actual change is SED_SD50 = 1.0 mm instead of 0.1 mm.	Same run configuration as case b except for output directory, sediment file name, plastic file name, and a shorter batch request on the cluster script.

4 Important Non-Obvious Differences

4.1 Case a is not a clean one-switch control for case b

At first glance, cases a and b look like a simple PLAST_FLOC off/on pair. They are not. In addition to enabling sediment and floc interaction in case b, the run namelists also change:

- STARTUP_TS_TYPE: observed in a, constant in b;
- RST_FIRST_OUT: day 58490.0 in a, day 58485.0 in b;
- NC_FIRST_OUT: day 58490.0 in a, day 58485.0 in b.

The startup file used by case a contains nontrivial vertical structure:

- salinity `ssl` ranges from about -0.0028 to 35.19 PSU;
- temperature `tsl` ranges from about 4.82 to 13.43°C .

By contrast, cases b and c start from spatially uniform constant $T = 20^\circ\text{C}$ and $S = 30$ PSU. Therefore case a differs from case b in both the plastic/sediment physics and the hydrodynamic initialization.

4.2 Case c only changes sediment grain size, not the plastic setup

The `generic_plastic_b.inp` and `generic_plastic_c.inp` files are identical. The only content difference recovered between b and c is:

SED_SD50 : 0.1 mm $\rightarrow$ 1.0 mm.

Other sediment parameters such as SED_WSET, SED_TAUE, and SED_TAUD remain unchanged in the provided files. So case c is best interpreted as a grain-size sensitivity test, not a broader sediment physics retuning.

4.3 No preserved model outputs are available in the case output folders

The directories OUTPUT_a, OUTPUT_b, and OUTPUT_c are presently empty. That means the original scenario intent has to be inferred entirely from the inputs and batch scripts; there is no saved output available here to confirm what diagnostics were originally examined.

5 Implications for the Next Mass-Conservation Test Design

For later diagnosis of `conc_a`/`conc_d` mass conservation, the recovered files suggest the following:

1. Case **a** is a useful broad baseline, but not a strict one-parameter control for case **b**, because the startup salinity/temperature treatment is also different.
2. Case **b** is the clearest representation of the intended flocc-enabled Delaware forcing case.
3. Case **c** is best interpreted as a sediment-size sensitivity around case **b**.
4. If the next objective is to isolate internal `conc_a`–`conc_d` conservation, it will be worth designing a cleaner pair of cases that keeps hydrodynamic initialization, output timing, and all plastic/sediment parameters fixed except the one switch or parameter being tested.

6 Supporting Scripts

Reusable inspection utilities have been placed in `FVCOM_MP_test_run/PYTHON`. The main script `inspect_delaware_cases.py` writes:

- `FVCOM_MP_test_run/PYTHON/output/case_inventory.json`;
- `FVCOM_MP_test_run/PYTHON/output/case_inventory.md`.

These files provide a machine-readable and human-readable summary of the same recovered scenario information.

7 Bottom Line

The recovered intent is most consistent with:

- **a**: no-floc baseline;
- **b**: flocc-enabled reference case;
- **c**: flocc-enabled sediment-size sensitivity case.

The main caution is that case **a** differs from case **b** in more than the flocc switch, so a later conservation campaign should probably build a cleaner control pair before interpreting mass-balance differences too aggressively.